

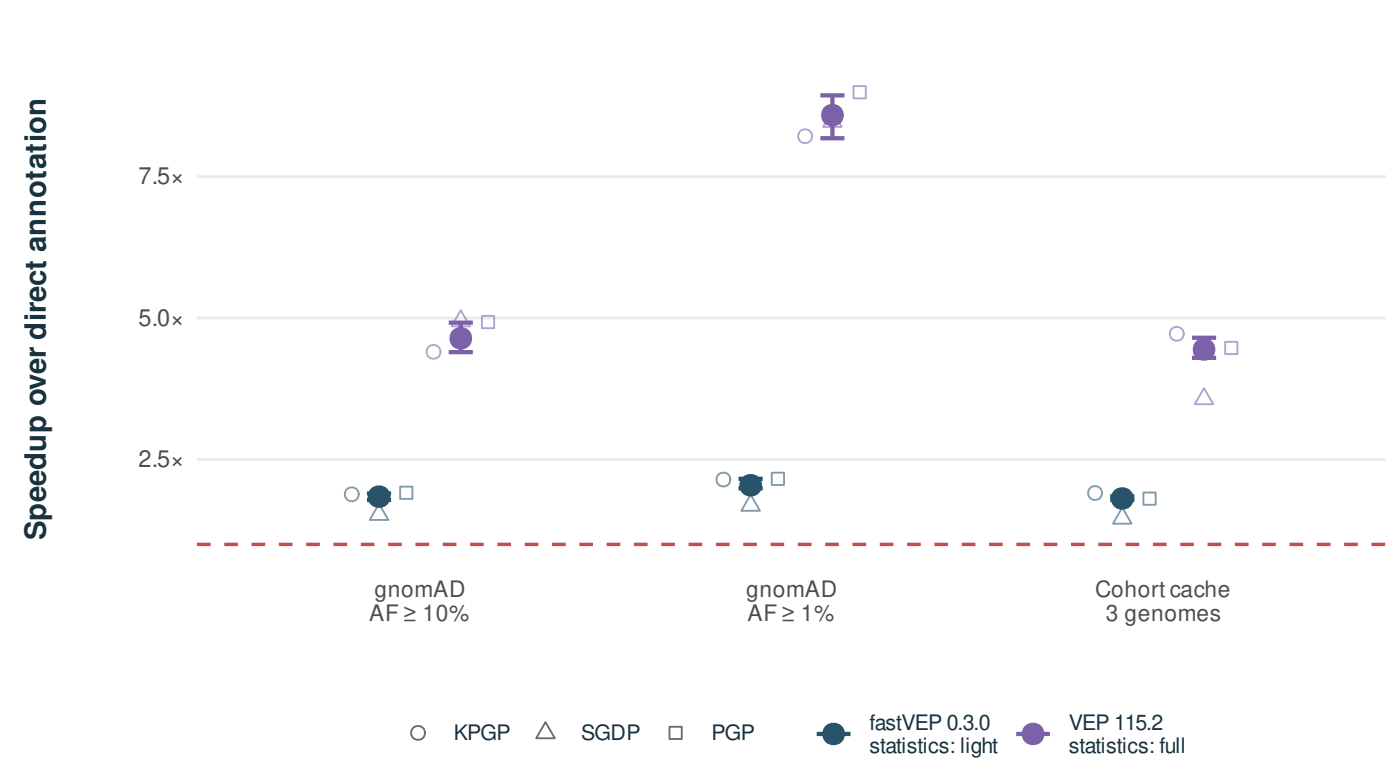
# VCFcache reduces runtime for both VEP and fastVEP

Exactly paired genomes, variants and cache blueprints · completed statistics sensitivity calibration

A

## Caching helps both annotators

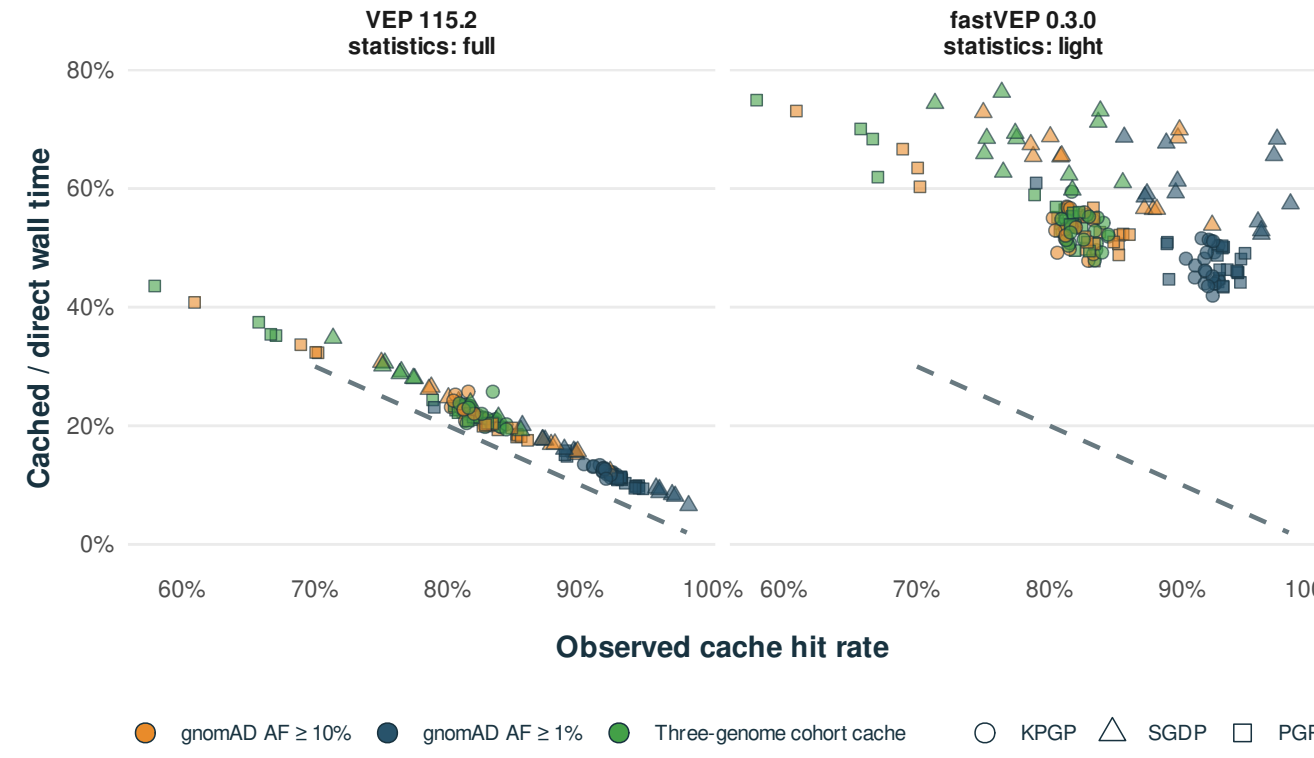
Large points: stratified median with 95% bootstrap interval; small symbols: cohort medians



B

## The same hits remove more work from the slower pipeline

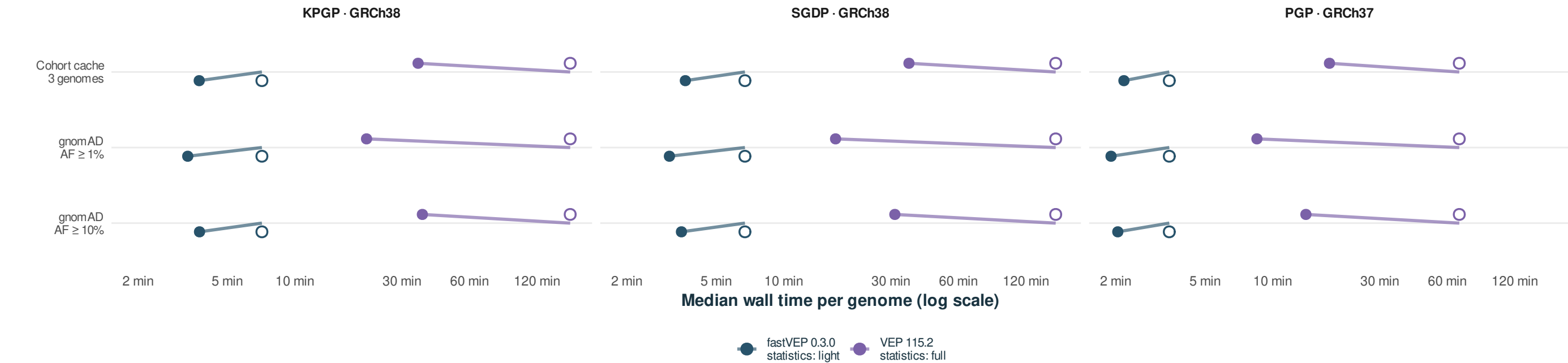
Dashed line: zero-overhead miss-fraction rule; axes are identical



C

## Absolute wall time spans minutes to hours

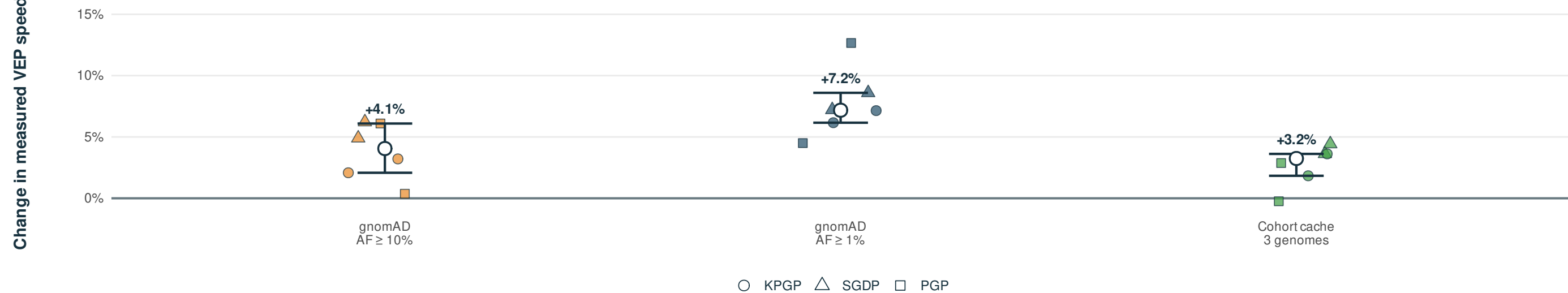
Open point: direct; filled point: cached; line: time returned per genome



D

## Light statistics make the historical VEP estimate slightly conservative

Six paired genomes (two per cohort); points compare light/full speedup, labels show medians



VEP 115.2 values are the measured --statistics full results; fastVEP 0.3.0 used --statistics light.  
The six-genome paired calibration is shown directly and is not used to rescale the 52 original VEP observations.  
Each genome/tool/strategy combination was run once; stratified bootstrap intervals treat genomes as inferential units.